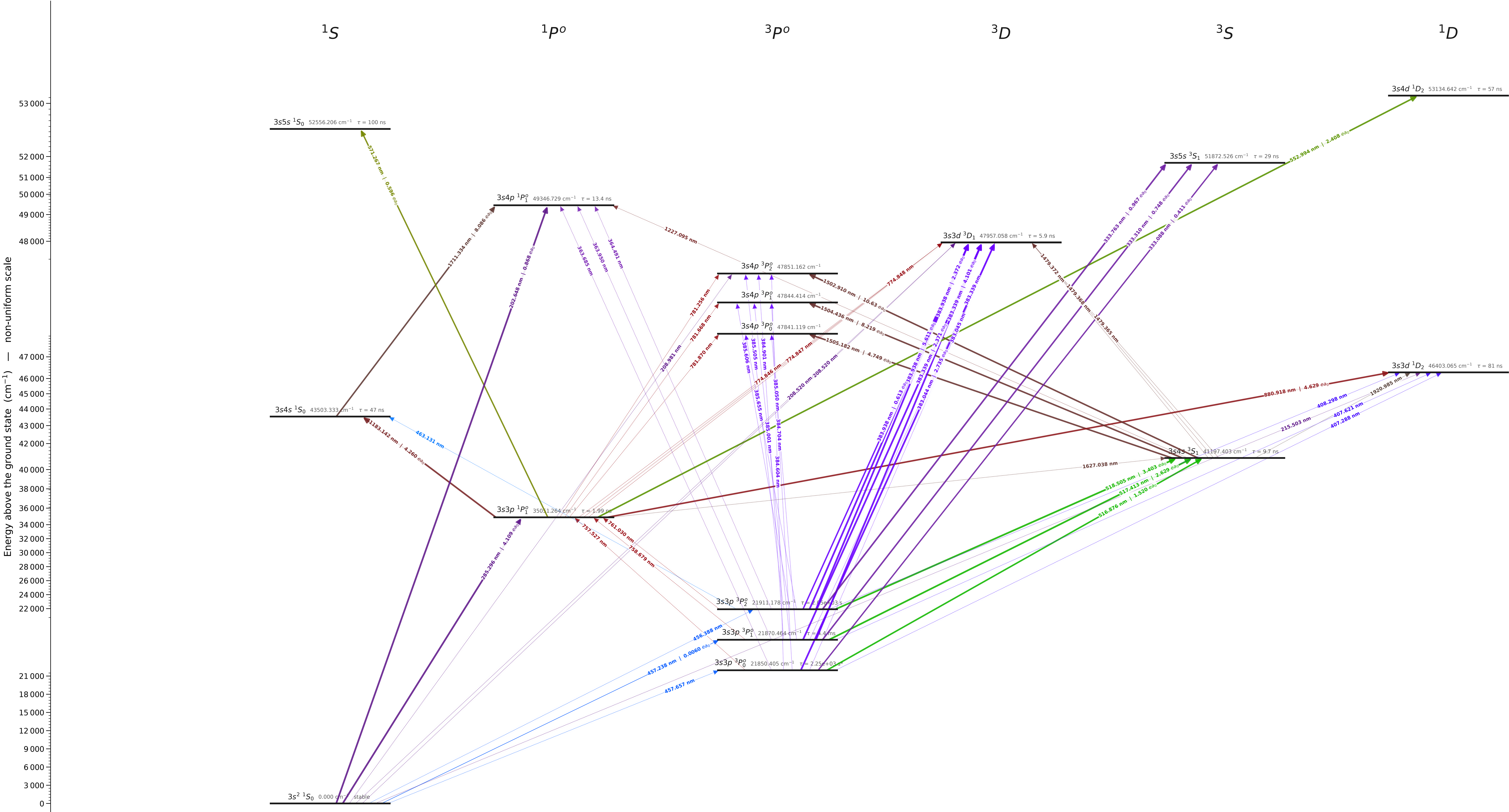
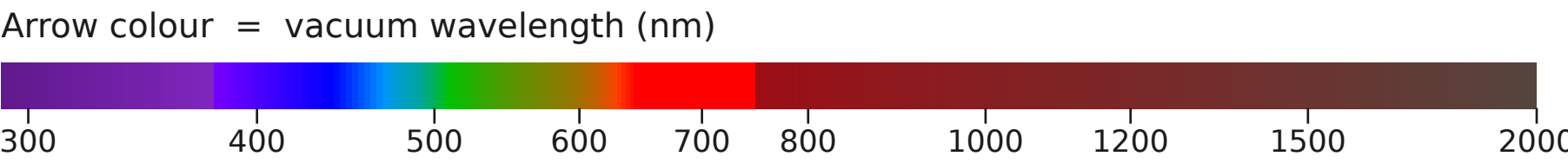


Magnesium-25

²⁵Mg (Mg I), nuclear spin *I* = 5/2. Energy levels and transitions below 2 μm: 18 levels, 61 lines.

Mg ⁺ ionisation limit 61671.05 cm⁻¹ = 7.64624 eV (162.151 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U\|er\|U\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U\ er\ U\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3s ² ¹S ₀ – 3s3p ¹P ₁ ^o	285.2964	285.2126	1050.8105	4.109	4.910e+08	79.98	97.7	singlet cooling / MOT / imaging line
3s3p ³P ₀ ^o – 3s3d ³D ₁	383.0441	382.9355	782.6578	2.735	8.990e+07	26.98	53	triplet MOT / repumping
3s3p ³P ₀ ^o – 3s3d ³D ₂	383.0446	382.9359	782.6568	–	2.110e-05	26.98	1.24e-11	triplet MOT / repumping
3s3p ³P ₁ ^o – 3s3d ³D ₁	383.3387	383.2299	782.0564	2.371	6.740e+07	26.98	39.8	triplet MOT / repumping
3s3p ³P ₁ ^o – 3s3d ³D ₃	383.3389	383.2301	782.0560	–	2.200e-05	26.98	1.3e-11	triplet MOT / repumping
3s3p ³P ₁ ^o – 3s3d ³D ₂	383.3391	383.2304	782.0555	4.101	1.210e+08	26.98	71.4	triplet MOT / repumping
3s3p ³P ₂ ^o – 3s3d ³D ₁	383.9379	383.8290	780.8358	0.613	4.480e+06	26.98	2.64	triplet MOT / repumping
3s3p ³P ₂ ^o – 3s3d ³D ₃	383.9381	383.8292	780.8355	5.611	1.610e+08	26.98	95	triplet MOT / repumping
3s3p ³P ₂ ^o – 3s3d ³D ₂	383.9383	383.8295	780.8349	2.372	4.030e+07	26.98	23.8	triplet MOT / repumping
3s ² ¹S ₀ – 3s3p ³P ₀ ^o	457.2377	457.1096	655.6600	0.006	2.540e+02	3.617e-05	100	intercombination line: narrow-line spectroscopy / Ramsey...
3s ² ¹S ₀ – 3s3p ³P ₀ ^o	457.6574	457.5292	655.0587	–	5.800e-04	7.074e-11	100	optical lattice clock transition
3s3p ³P ₀ ^o – 3s4s ³S ₁	516.8761	516.7321	580.0084	1.520	1.130e+07	16.41	11.2	triplet optical pumping / repumping, metastable-beam spec...
3s3p ³P ₁ ^o – 3s4s ³S ₁	517.4125	517.2684	579.4071	2.629	3.370e+07	16.41	33.9	triplet optical pumping / repumping, metastable-beam spec...
3s3p ³P ₂ ^o – 3s4s ³S ₁	518.5048	518.3604	578.1865	3.403	5.610e+07	16.41	54.9	triplet optical pumping / repumping, metastable-beam spec...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁵Mg hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3s3p ³P ₁ ^o	-144.977(5)	-8.308(5)	F=3/2: +501.60 F=5/2: +151.62 F=7/2: -364.52
3s3p ³P ₂ ^o	-128.445(5)	16.009(5)	F=1/2: +910.32 F=3/2: +710.45 F=5/2: +381.33 F=7/2: -71.03 F=9/2: -638.22
3s4s ³S ₁	-321.6(15)	–	F=3/2: +1125.60 F=5/2: +321.60 F=7/2: -804.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3s ² ¹S ₀ – 3s4p ¹P ₁ ^o	202.648	26-24	2918.9(16)
3s ² ¹S ₀ – 3s4p ¹P ₁ ^o	202.648	25-24	1526.4(16)
3s ² ¹S ₀ – 3s3p ¹P ₁ ^o	285.296	26-24	1414(8)
3s ² ¹S ₀ – 3s3p ¹P ₁ ^o	285.296	25-24	741(9)
3s3p ³P ₀ ^o – 3s3d ³D ₁	383.044	26-24	60(15)
3s3p ³P ₀ ^o – 3s3d ³D ₂	383.045	26-24	60(15)
3s3p ³P ₁ ^o – 3s3d ³D ₁	383.339	26-24	61(3)
3s3p ³P ₁ ^o – 3s3d ³D ₃	383.339	26-24	61(3)
3s3p ³P ₁ ^o – 3s3d ³D ₂	383.339	26-24	61(3)
3s3p ³P ₂ ^o – 3s3d ³D ₁	383.938	26-24	58(4)
3s3p ³P ₂ ^o – 3s3d ³D ₃	383.938	26-24	58(4)
3s3p ³P ₂ ^o – 3s3d ³D ₂	383.938	26-24	58(4)
3s ² ¹S ₀ – 3s3p ³P ₀ ^o	457.238	26-24	2683.180(20)
3s ² ¹S ₀ – 3s3p ³P ₁ ^o	457.238	25-24	1405.20(10)
3s ² ¹S ₀ – 3s3p ³P ₀ ^o	457.657	26-24	2681.467(20)
3s ² ¹S ₀ – 3s3p ³P ₀ ^o	457.657	25-24	1403.90(10)
3s3p ³P ₀ ^o – 3s4s ³S ₁	516.876	26-24	-391.3(17)
3s3p ³P ₀ ^o – 3s4s ³S ₁	516.876	25-24	-205.7(15)
3s3p ³P ₁ ^o – 3s4s ³S ₁	517.413	26-24	-390.1(14)
3s3p ³P ₁ ^o – 3s4s ³S ₁	517.413	25-24	-209.1(13)
3s3p ³P ₂ ^o – 3s4s ³S ₁	518.505	26-24	-394.4(8)
3s3p ³P ₂ ^o – 3s4s ³S ₁	518.505	25-24	-205.7(8)

Shift = ν(first isotope) – ν(second isotope).